



Practical Design and Training of Edge Deployable VQA in a Natural Interaction Generative Edge Pipeline

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System Research and Applications

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Agenda

Introduction

System Overview

VQA Model Development

Model Performance

System Deployment

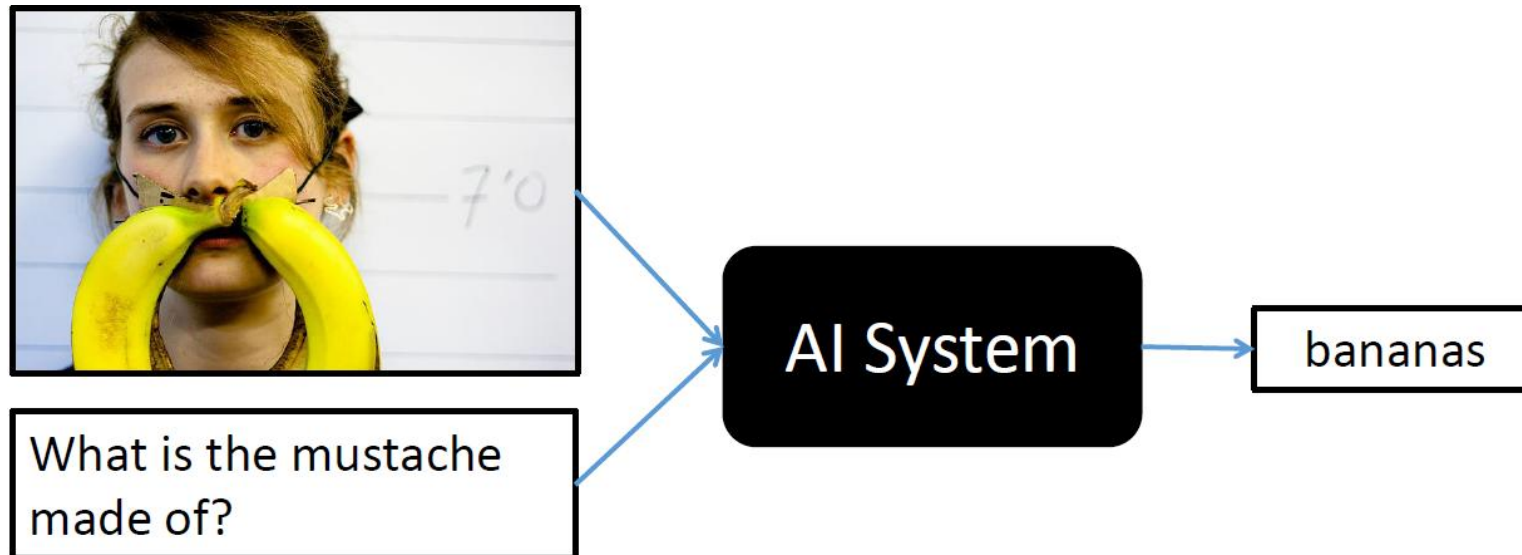
Conclusions

Introduction



Visual Question Answering (VQA) for EdgeAI

- Input: image + text question → Output: predicted answer.
- Provides a **concrete case study** and learn how to develop an advanced AI workload for an Edge processor.





Approach and Contributions

A Practical tutorial: building lightweight VQA models optimized for STM32MP2 exploiting the embedded NPU.

- Practical, reproducible by everyone, designed for students and young professionals.

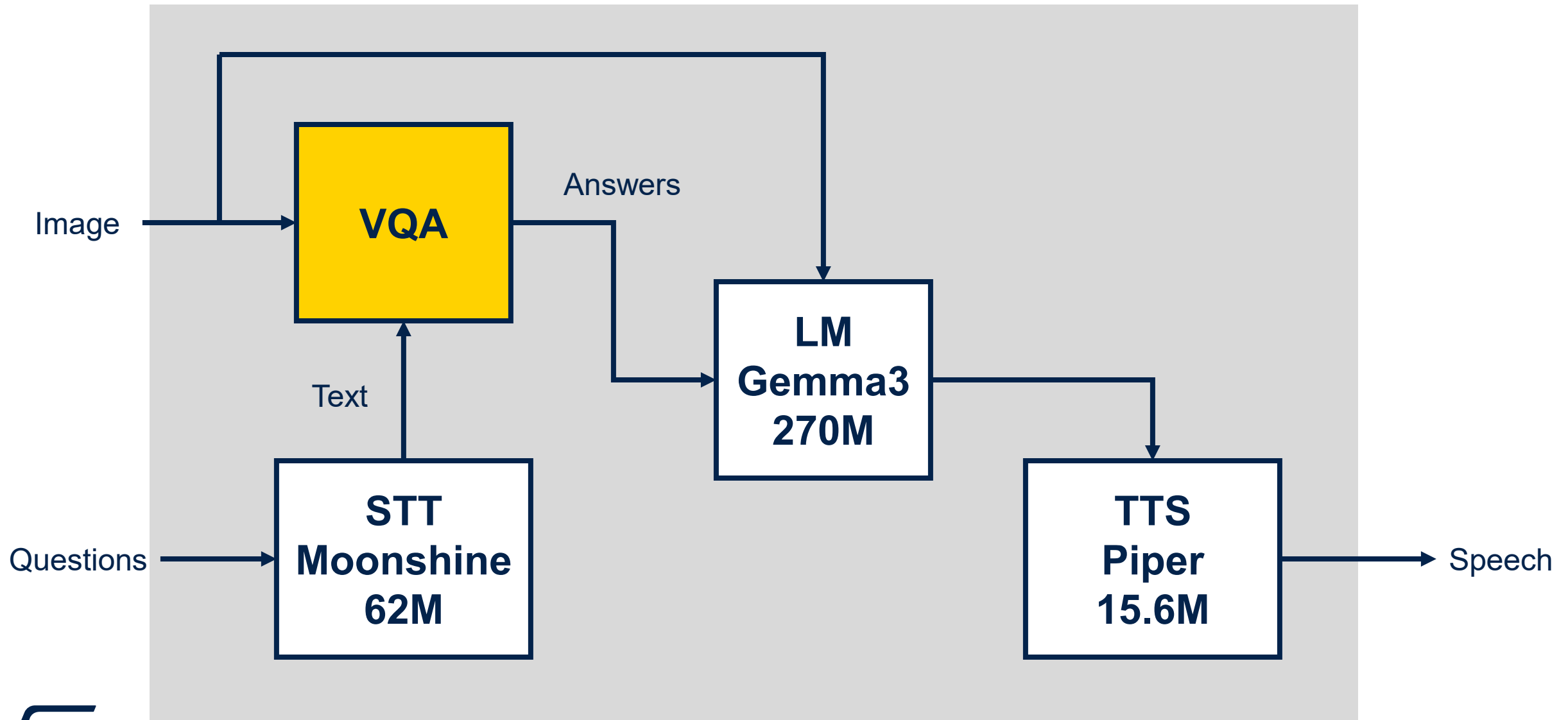
Help to build a modular generative edge system: given an image and a spoken question, generates a natural spoken answer directly on edge hardware

- Speech-to-Text (SST) - github
- **Lightweight VQA model** - learn from this tutorial
- Edge Language Model (ELM) – Foundational model
- Text-To-Speech (TTS) - github

System Overview



Multi modal Agent



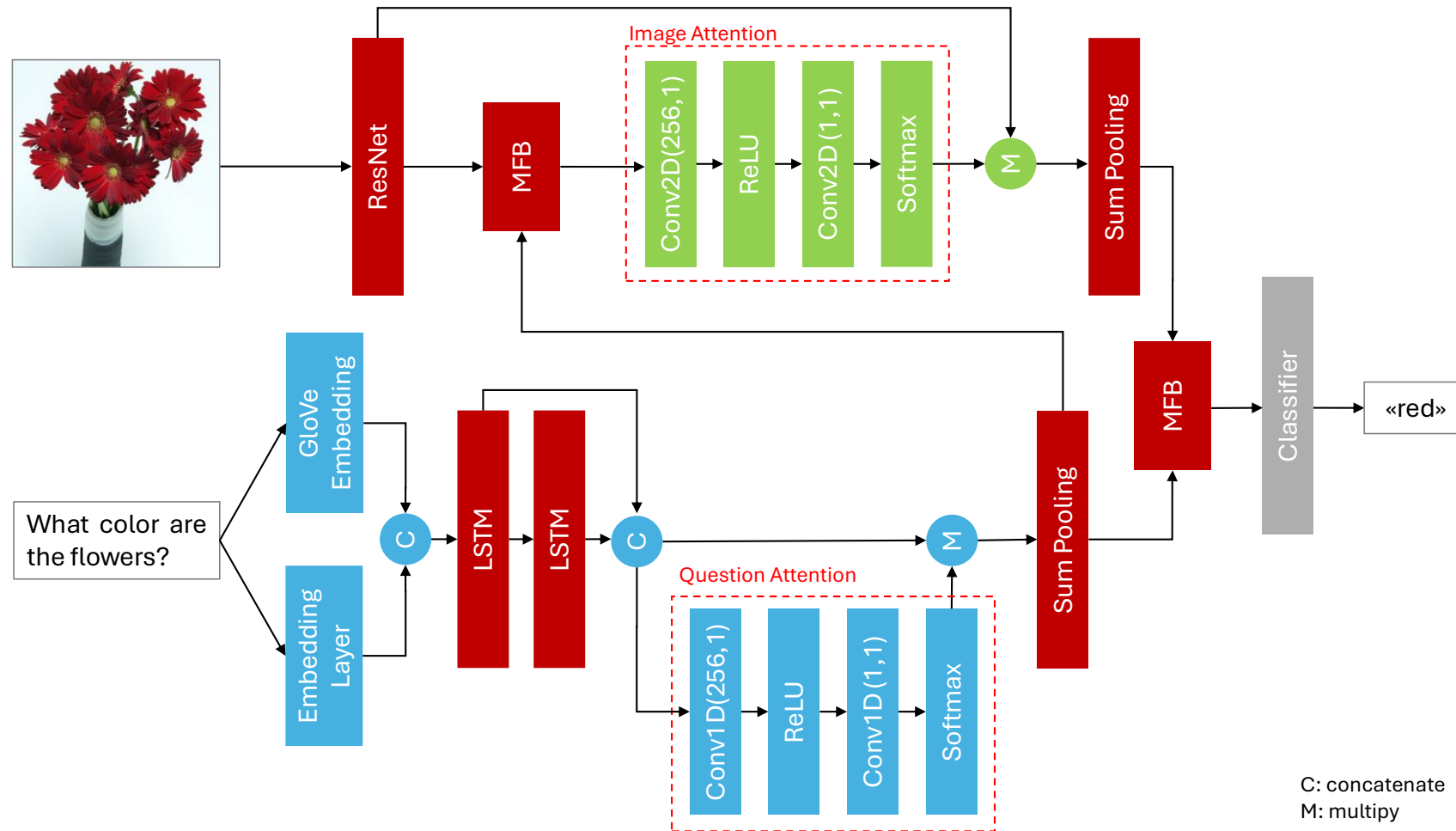
VQA Unfriendly Edge NPU Model



VQA Model Architectures

Architectures based on Yu et al. [1]:

- MFB Baseline (no attention layers)
- MFB Attention (only image attention layer)
- MFB Co-Attention (both image and question attention layers)
- Feature extraction:
 - Text $\rightarrow$ LSTM
 - Image $\rightarrow$ ResNet-152

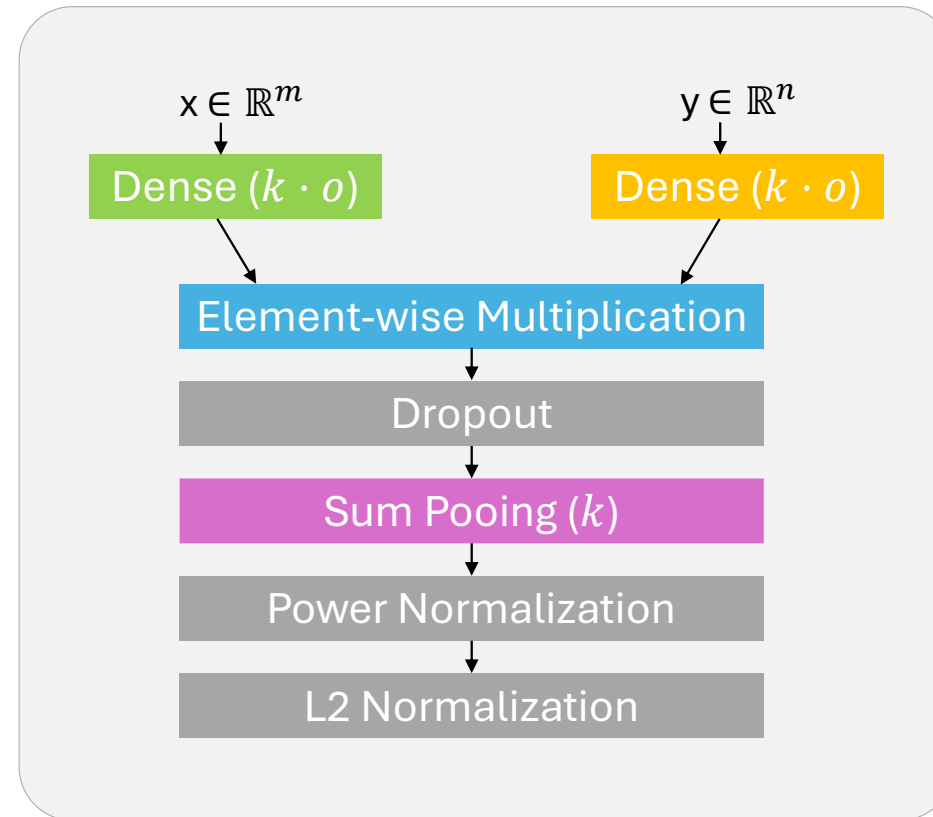


VQA Model Architectures

MFB module

Architectures based on Yu et al. [1]:

- Fusion: Multimodal Factorized Bilinear (MFB) module
- k and o are hyperparameters of the module



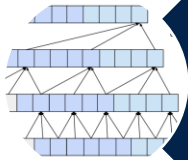
VQA Friendly Edge NPU Model



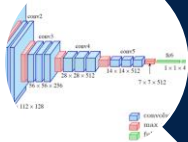
Optimizations STM32MP2 friendly



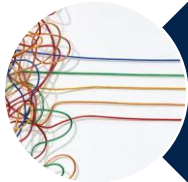
Optimizations aimed to achieve efficient execution with/without STM32MP2 NPU acceleration



LSTM replaced by Temporal Convolutions (TCLs)
For faster, hardware-friendly sequence modeling.



ResNet-152 (60.3M) replaced by MobileNetV3 Large (5.5M) more suitable for edge devices.



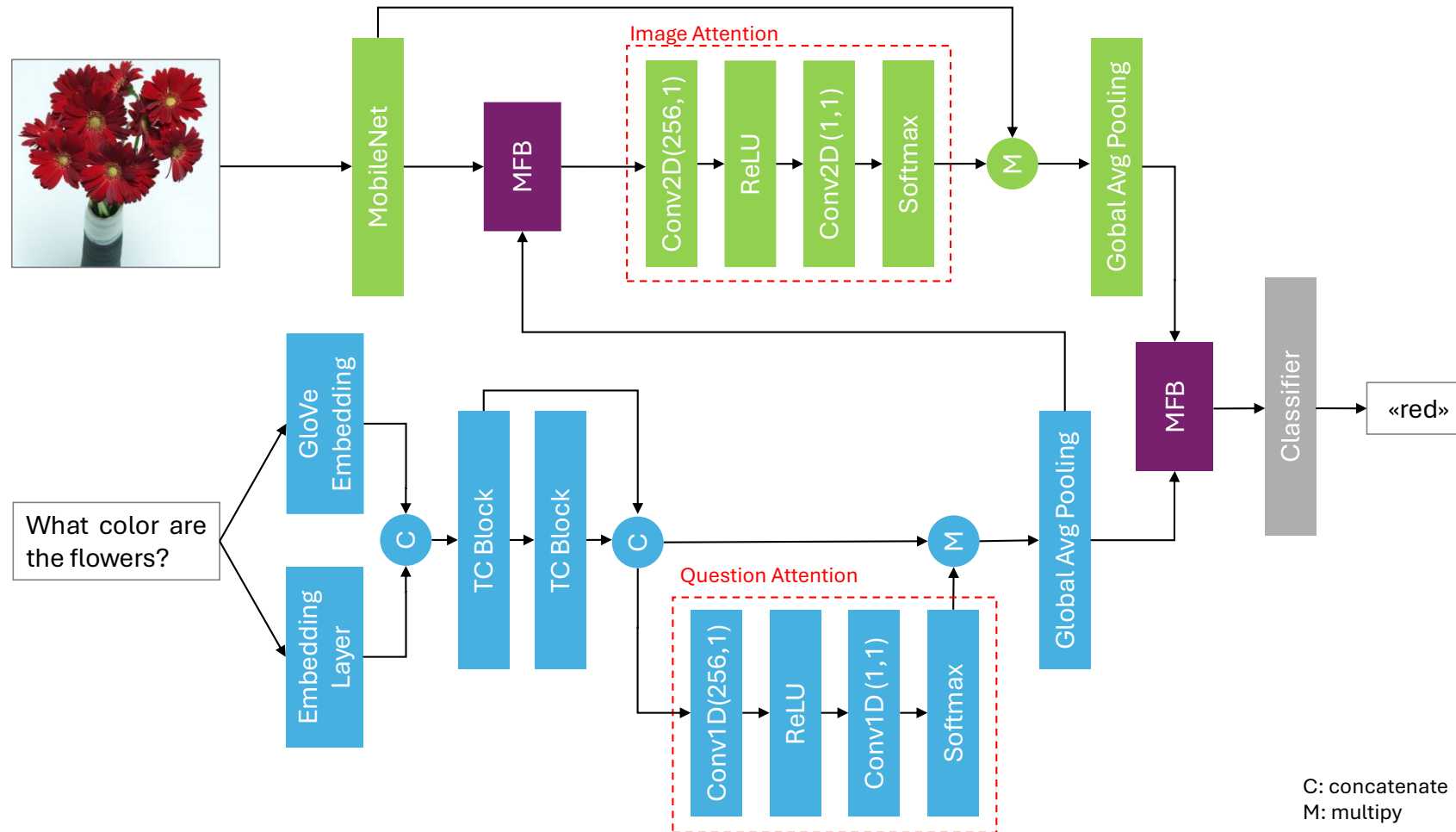
MFB simplifications: average pooling, removed power normalization



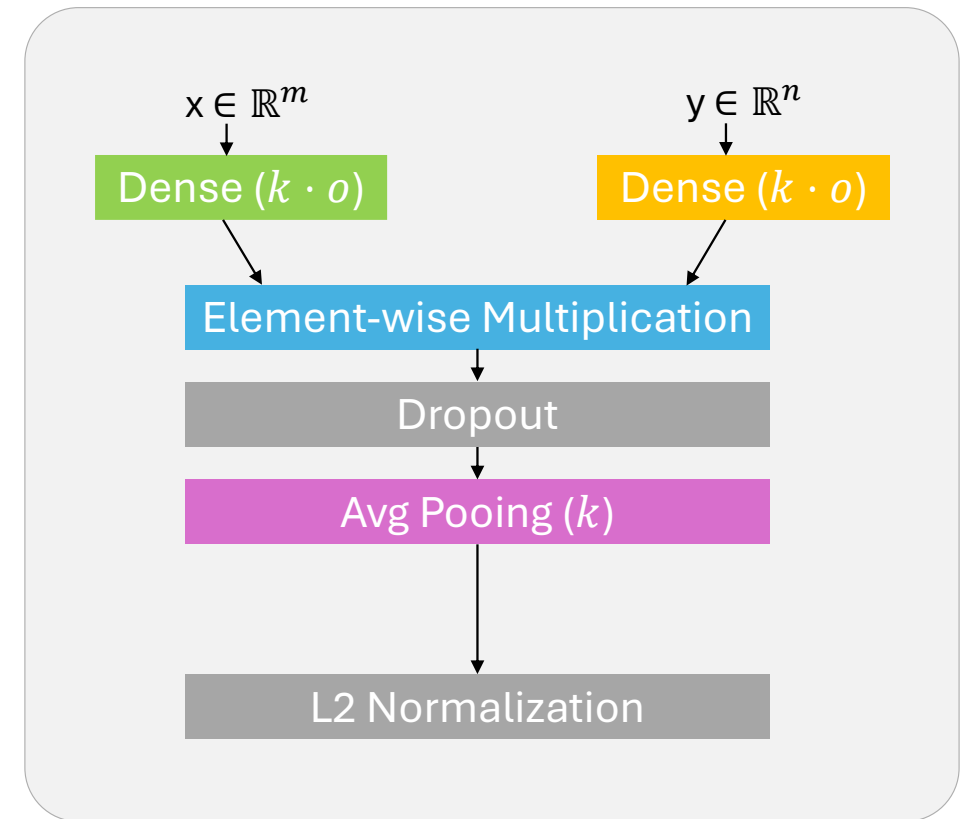
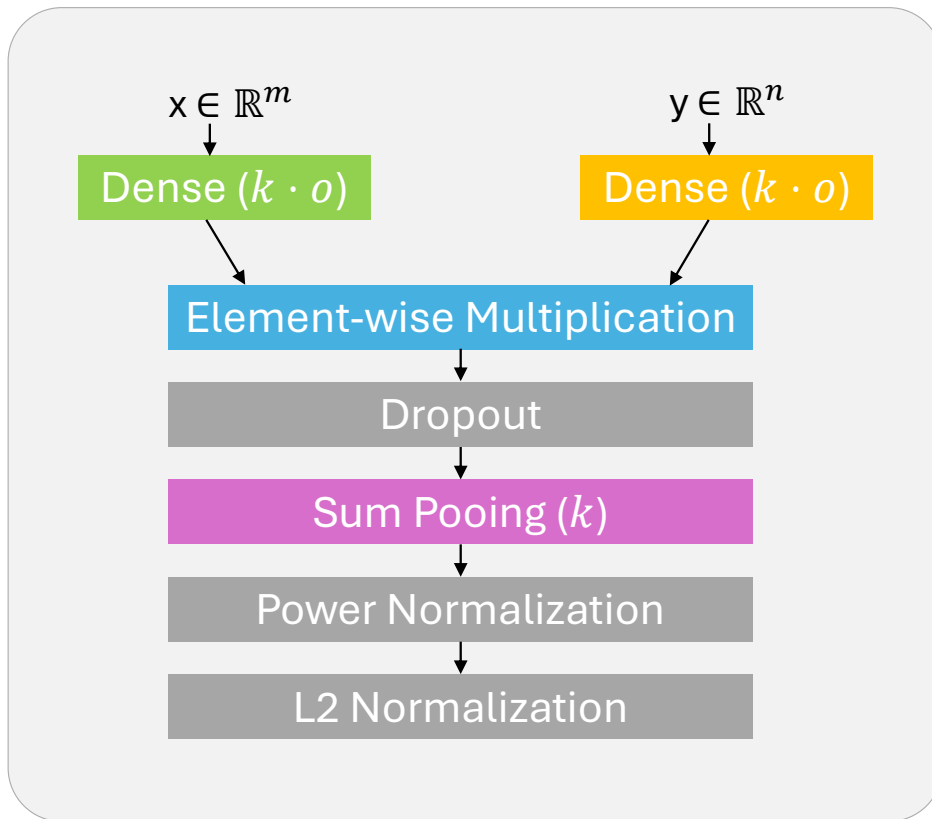
Word embeddings: concatenation of learned embeddings with pre-trained GloVe for richer semantics.

Optimized VQA Architectures

- MFB CoAttention

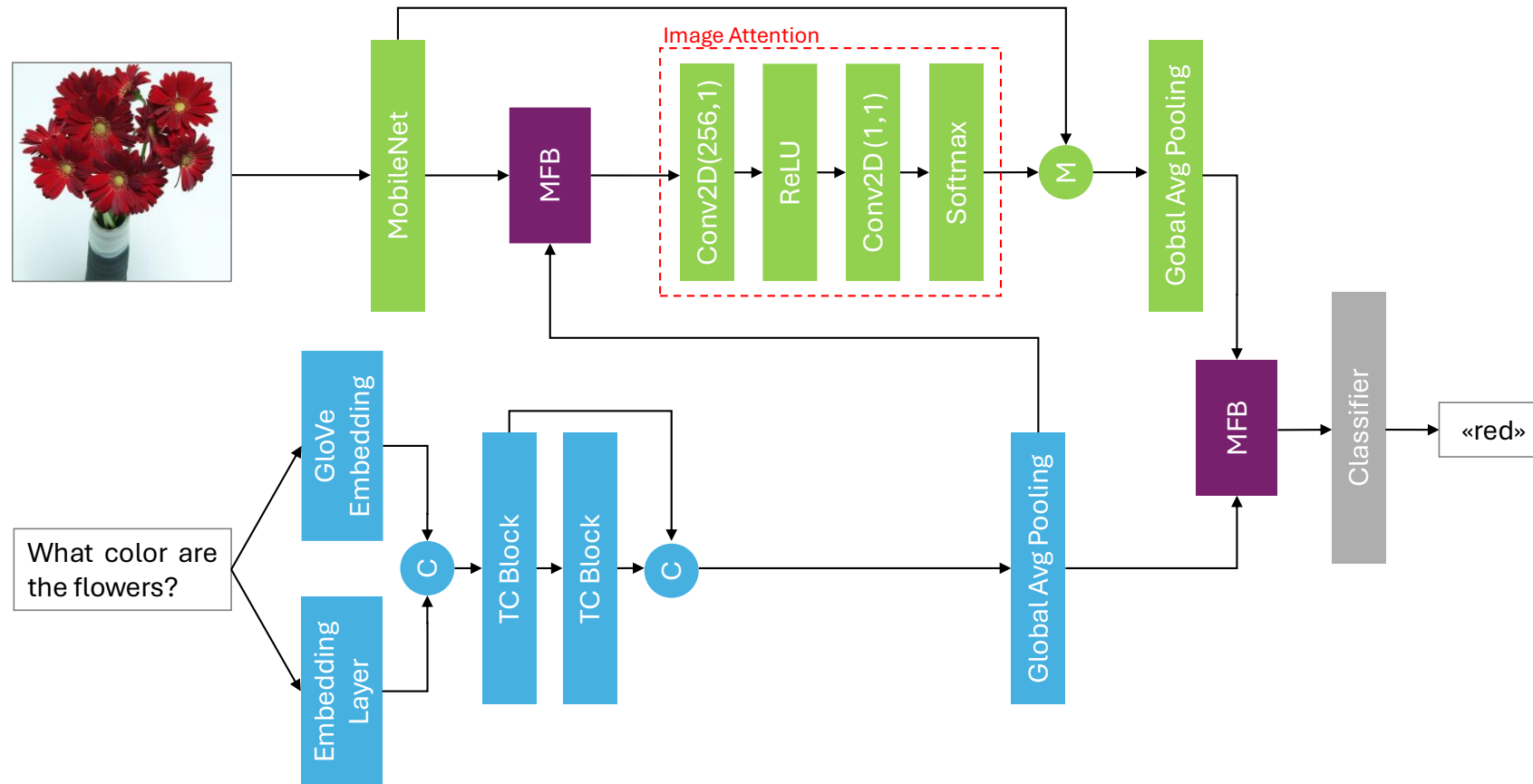


Optimized MFB



Optimized VQA Architectures

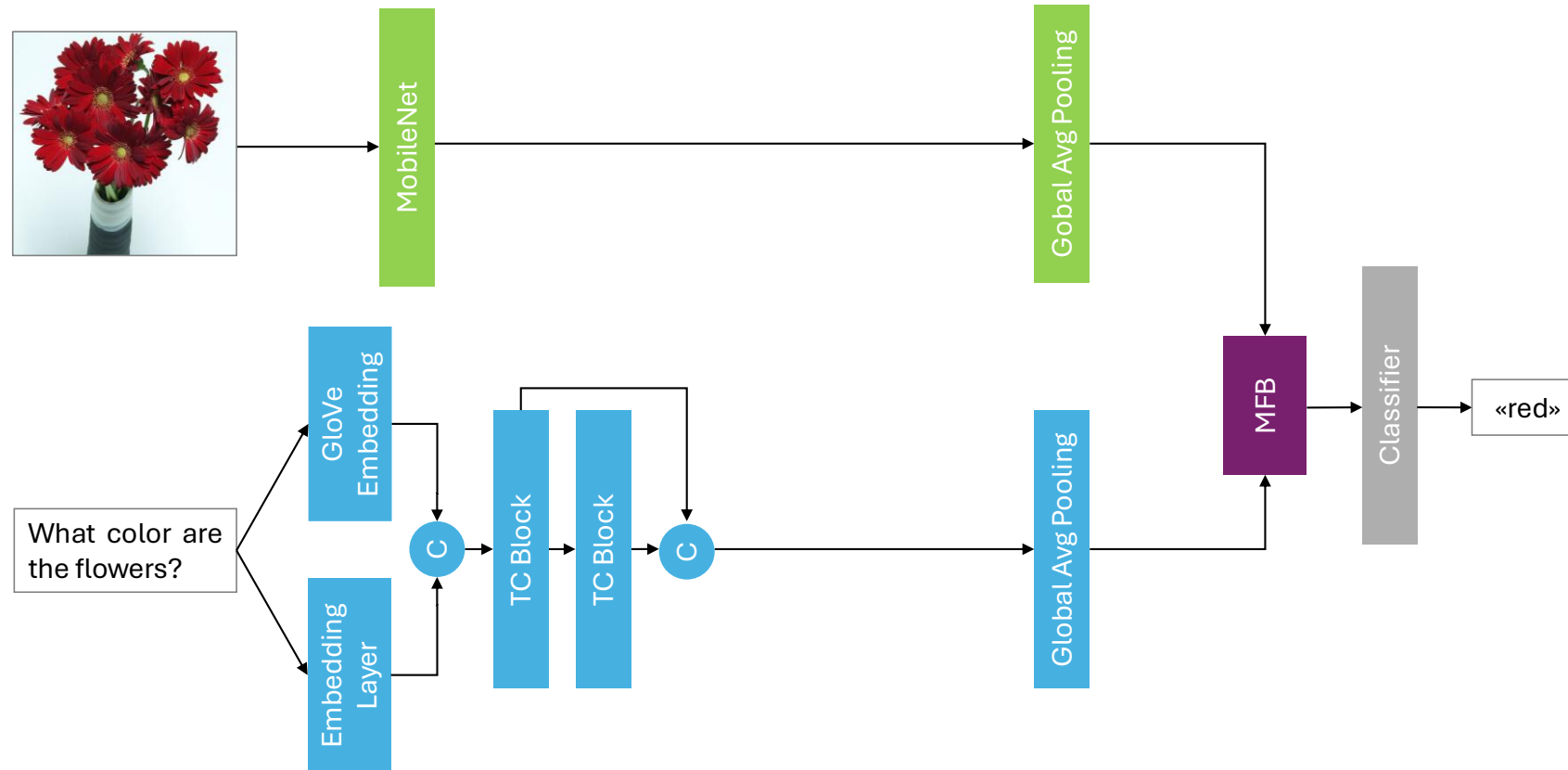
- MFB Attention



C: concatenate
M: multiply

Optimized VQA Architectures

- MFB Baseline



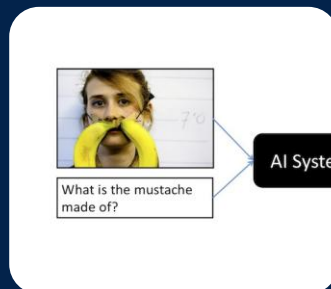
C: concatenate
M: multiply

Dataset





Dataset and Processing



Dataset: VQA v2

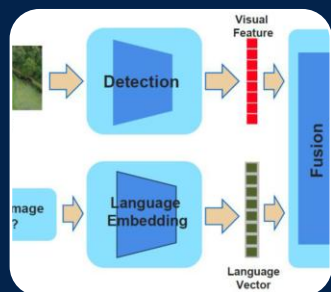
~5 questions per image

1 ground truth answer + 10 human answers per question

Pre-split into **train / val**

Evaluation performed on **validation set**

Set	Questions	Images
Training	443,757	82,783
Validation	214,354	40,504

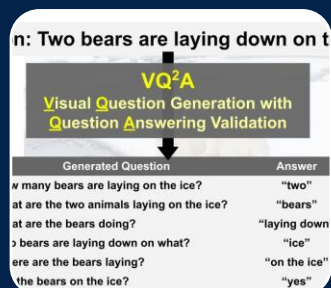


Processing:

Answers: normalized (official preprocessing guidelines)

Images: resized to 224x224, rescaled to [-1,1]

Questions: tokenized (vocabulary: 6415 tokens), padded/truncated to 15 tokens

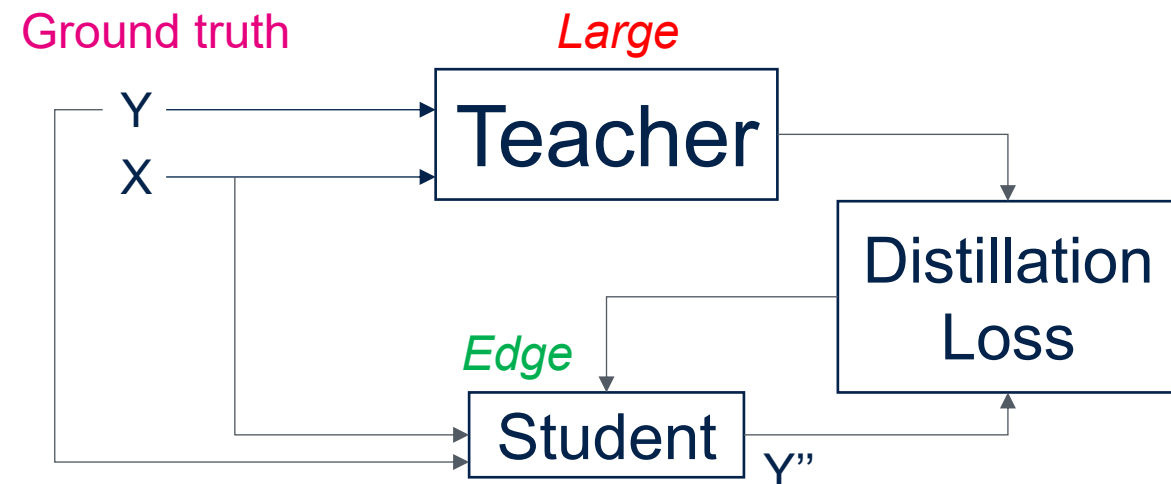


Answer space: top 1,000 most frequent answers

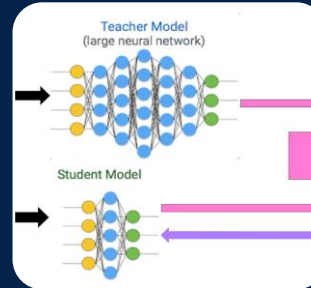
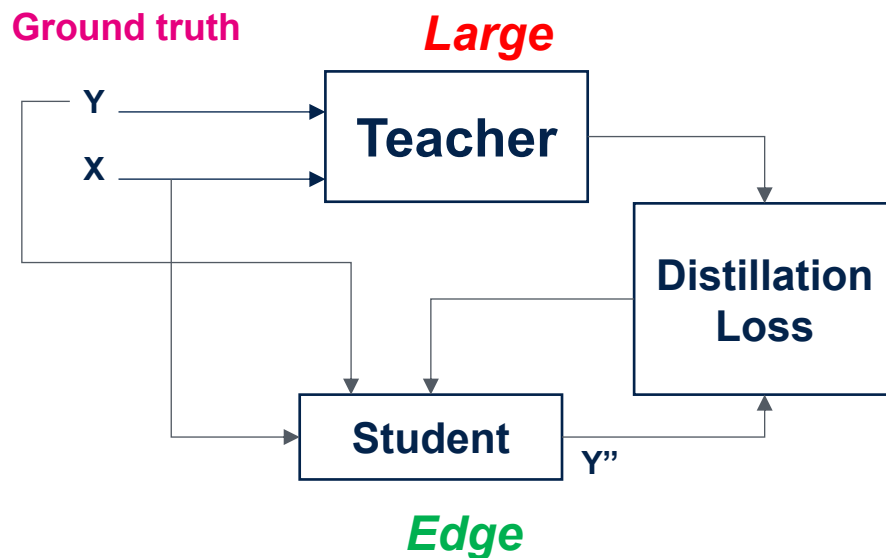
Training set reduced to 87.5% of original size

Training Procedure

- **Knowledge Distillation (KD)**
 - Student achieves higher accuracy than training from scratch
 - Method of Hinton et al. [2] (logit-based KD): combines cross-entropy (ground truth) and distillation loss (teacher's softened predictions)
- **Teacher model: BEiT-3** fine-tuned on VQAv2
 - Accuracy: **82.53%** (test-dev split)
 - 3129 possible answers
 - **683M parameters**
- **Improvements:**
 - **Balanced loss** to handle dataset imbalance and prevent overfitting to the most frequent answers



Training Procedure



Knowledge Distillation (KD)

Student achieves higher accuracy than training from scratch

Method of Hinton et al. [2] (logit-based KD): combines cross-entropy (ground truth) and distillation loss (teacher's softened predictions)



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683M parameters



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Model Performance



Model Performance

- **Evaluation metric:**

- **Accuracy** = agreement between model's predicted answer and **human annotators**

$$accuracy = \frac{1}{N} \sum_i \min \left(\frac{count(a_i)}{3}, 1 \right)$$

- Model's predicted answer a_i for question i
- Number of questions N
- $count(a_i)$ number of annotators who provided the answer a_i to question i

Model	#Params	Accuracy
MFB (original)	68.0 M	65.4%
MFB base.	24.4 M	56.0%
MFB att.	40.4 M	57.0%
MFB co-att.	51.2 M	56.4%

Top-10 ans. (highest accuracy)	Ans. 11-50	Ans. 51-1000
92.1%	85.1%	63.1%

Average accuracy of MFB att., computed by ranking the 1000 possible answers by their per-answer accuracy.”

Model Performance

- **Models vs. Original MFB and BEiT-3:**
 - Models achieve lower accuracy than original MFB model (65.4%) and SOTA
 - Also lower accuracy than teacher model BEiT-3 (87.5%)
 - Expected due to basic processing and simplified training procedure
- **Trade-off:** fewer parameters and efficient on STM32MP2 but reduced accuracy
- **Best results: MFB att.** → integrated into the generative system and **benchmarked on STM32MP2 (NPU)**

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System Deployment



VQA Acceleration on STM32MP2



Benefits of NPU acceleration:

Reduced runtime (~13.56× speed-up)

Reduced power consumption

Models (this tutorial)	Execution Device	Inference time [ms]	NPU usage %
MFB att.	GPU/NPU	17.8	96
MFB att.	2 A35 CPUs	241.41	0

Use case analysis on STM32

- Runtime (RTF/inference time/throughput), average power consumption, memory footprint
- $RTF = \frac{\text{processing time}}{\text{audio lenght}}$

Model	Framework	Execution Device	Runtime Measure	Power (W)	Flash (MB)	RAM (MB)
Moonshine	Onnx runtime	CPU	Real time factor: 1.50	0.89	244.8	1101
MFB att. (tutorial)*	This work	GPU/NPU	Inference time: 17.8 ms	0.75	46.9	69.6
Gemma 3 270M	Ollama	CPU	Throughput (tok./s): 4.07	1.00	292	789
Piper	Onnx runtime	CPU	Real time factor: 1.84	0.81	61	356



Visual Question Answering (VQA) on Edge Devices

1. Introduction

This repository provides a hands-on tutorial that shows you how to design, train, and evaluate **Visual Question Answering (VQA)** models for **resource-constrained edge devices**, specifically the **STM32MP2 MPU**.

VQA is a task at the intersection of computer vision and natural language processing: given an **image** and a **natural language question** about it, the model must generate the correct **answer**. VQA has many real-world applications, such as accessibility tools for visually impaired users and smart assistants that understand visual content. However, most state-of-the-art VQA models are **large and resource-hungry**, making them unsuitable for **edge devices**.

The models in this tutorial are implemented in Keras 3.9, optimized for TensorFlow Lite, and designed to leverage the STM32MP2 NPU for efficient inference.

This README is structured as follows:

- [Section 2](#): Quickstart (inference in 3 steps)
- [Section 3](#): Tutorial (models, dataset, training and evaluation)
- [Section 4](#): Usage Guide
- [Section 5](#): Performance Testing on STM32MP2
- [Section 6](#): Configuration and Advanced Options



Tutorial:

Guide to develop lightweight multimodal VQA models
Deployable and accelerated on STM32MP2

Proof-Of-Concept Generative System:

Combines VQA + LM + STT + TTS

Enables multimodal and natural interaction on edge devices

Evaluated for latency, energy and memory on STM32MP2

Future Directions:

Build agents on STM32MP2

Conclusions

VQA model

“What is the boy doing?”



→ “Writing”

Generative System

🔊 “What is the boy doing?”



🔊 → “He is writing a summary of yesterday math lesson”

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